

How a normal machine-learning model predicts EPIT

Conventional machine learning

- A conventional model is trained directly on the available corrosion dataset.
- It repeatedly compares predictions with measured EPIT values and adjusts its parameters.
- The result is one model fitted specifically to this dataset.
- A different target or dataset normally requires new training.
- With limited and heterogeneous corrosion data, a flexible model can overfit dataset-specific patterns.

A model pretrained to learn from tables

Tabular in-context foundation models

- TabICL is pretrained before seeing the real EPIT task.
- Pretraining uses many automatically generated synthetic tables.
- Every table contains example rows and rows whose targets must be predicted.
- During EPIT evaluation, measured training rows are supplied as context.
- The pretrained parameters remain fixed.

The model still needs labeled EPIT examples. It uses them as context during prediction rather than to retrain its parameters.

The corrosion task

- **Source:** Nyby et al., *Scientific Data* (2021); an open-access Figshare dataset compiled from 85 literature sources.
- The pitting-potential section contains **810** measurements across Fe-, Al-, Ni-Cr-Mo, high-entropy and other alloys; **760** have a usable numeric average EPIT.
- **21 input features:**
 - 17 alloy-composition columns;
 - temperature, chloride concentration, and pH;
 - test method.
- Higher EPIT generally indicates stronger resistance to pit initiation under comparable conditions.
- EPIT depends jointly on alloy chemistry, environment, and experimental procedure.
- The dataset combines measurements from different test protocols.

	A	B	C	D	E	F	G	H	I
1	Composition, wt.% Fe	Composition, wt.% Cr	Composition, wt.% Ni	...	Test Temp. oC	[Cl-] M	[Cl-] pH	[Cl-] Test Method	Epit, mV (SCE) Avg.
2									
3	69.705	18	10		50	0.005133497	7.8	Potentiodynamic Polarization wit	287
4	69.705	18	10		50	0.008555828	7.8	Potentiodynamic Polarization wit	260
5	69.705	18	10		50	0.017111655	7.8	Potentiodynamic Polarization wit	218
6	69.705	18	10		50	0.085558276	7.8	Potentiodynamic Polarization wit	140
7	69.705	18	10		50	0.546	8.2	Potentiodynamic Polarization wit	94
8	67.535	16	11		50	0.005133497	7.8	Potentiodynamic Polarization wit	344
9	...								
10	69.819	18.92	8.51		25	0.5	-0.301	Potentiodynamic Polarization	265.5
11	69.595	18.87	8.59		25	0.5	-0.301	Potentiodynamic Polarization	328.5
12	69.601	18.8	8.63		25	0.5	-0.301	Potentiodynamic Polarization	532.3
13	58.389	17.38	20.15		25	0.6	7.68	Following pretreatment for 5 min	-65
14	57.048	17.68	20.43		25	0.6	7.68	Following pretreatment for 5 min	-25
15	...								
16	68.534	16.3	14.4		60	0.1	7	Polarization curves were obtaine	77.7
17	66.475	16.3	14		60	0.1	7	Polarization curves were obtaine	214.3
18	63.874	16.5	14.1		60	0.1	7	Polarization curves were obtaine	297.1
19	66.475	16.3	14		60	0.1	1	Polarization curves were obtaine	58.8377724
20	63.874	16.5	14.1		60	0.1	1	Polarization curves were obtaine	121.3075061
21	60.344	16.17	14.1		60	0.1	1	Polarization curves were obtaine	208.4745763
22	56.644	16.22	14		60	0.1	1	Polarization curves were obtaine	252.0581114
23	66.475	16.3	14		60	0.1	1	Polarization curves were obtaine	122.0007624

Performance: conventional ML vs. in-context models

- CatBoost is trained on each corrosion training split, whereas TabICL uses the same labeled rows as context without updating its pretrained parameters.
- Using the same five data splits and evaluation metrics, pretrained TabICL V2 provides the strongest overall performance.

Model	Spearman	MAE (mV)	RMSE (mV)	R^2
CatBoost	0.8300	144.7	216.6	0.7395
Generic retrained TabICL	0.8372	151.4	212.7	0.7481
Pretrained TabICL V2	0.8528	125.6	207.7	0.7592

The case for an informed prior

- Generic TabICL sees anonymous columns such as $X_1, X_2, \dots$
- It does not know which feature represents chromium, chloride, temperature, or test method.
- Its synthetic target has no pitting-potential interpretation.
- It can learn general statistical patterns, but its synthetic experience contains no deliberate corrosion structure.

Central project idea: Improve TabICL by redesigning some synthetic practice problems to resemble the organization of the EPIT task.

Corrosion-informed feature generation

Physical feature transformations

For corrosion-informed tasks:

- The 17 material positions are converted into nonnegative composition-like values.
- Softmax or masked Dirichlet generation controls how dense or sparse the alloy is.
- Environment positions become temperature-, chloride-, and pH-like variables.
- The process position becomes one of 52 synthetic test-method categories.
- Feature order remains fixed so the target mechanism can interpret the correct positions.

PREN-inspired synthetic EPIT target

$$p = \text{std}(\text{Cr} + 0.25 \text{Ni} + 3.3 \text{Mo} + 1.65 \text{W})$$

$$P = \tanh(p), \quad S = \sigma(-p), \quad C = \sigma(\text{std}(\log_{10}[\text{Cl}^-]))$$

$$E = \sigma(\text{std}[w_T T + w_{\text{Cl}} \log_{10}[\text{Cl}^-] + w_{\text{pH}} |\text{pH} - \text{pH}_{\text{neutral}}|])$$

$$g_{\text{EPIT}} = c_{\text{mat}} P - c_{\text{env}} E - c_{\text{int}} S C + c_{\text{proc}} M$$

- P : passivity; c_{mat} raises EPIT for more resistant compositions.
- E : environmental aggressiveness; c_{env} lowers EPIT under harsher conditions.
- S : alloy susceptibility, which increases when passivity is low.
- C : chloride severity; c_{int} increases the penalty for susceptible alloys.
- M : test-method offset; c_{proc} represents protocol-dependent shifts.

$$y_{\text{synthetic}} = (1 - \lambda) z_{\text{generic}} + \lambda z_{\text{EPIT}}$$

λ controls the balance between generic and corrosion-informed target structure.

Direct analysis of the PREN-inspired target

Before transformer training, **1,024 sampled PREN-inspired target rules** were applied directly to the real alloy and environmental features.

Their average output was compared with the observed EPIT ranking.

$$\text{Spearman correlation} = \mathbf{0.4929}$$

This positive correlation shows that the designed material, environmental, and chloride-interaction terms broadly reproduce the EPIT ordering observed in the real data.

Adding alloy-level chemical information

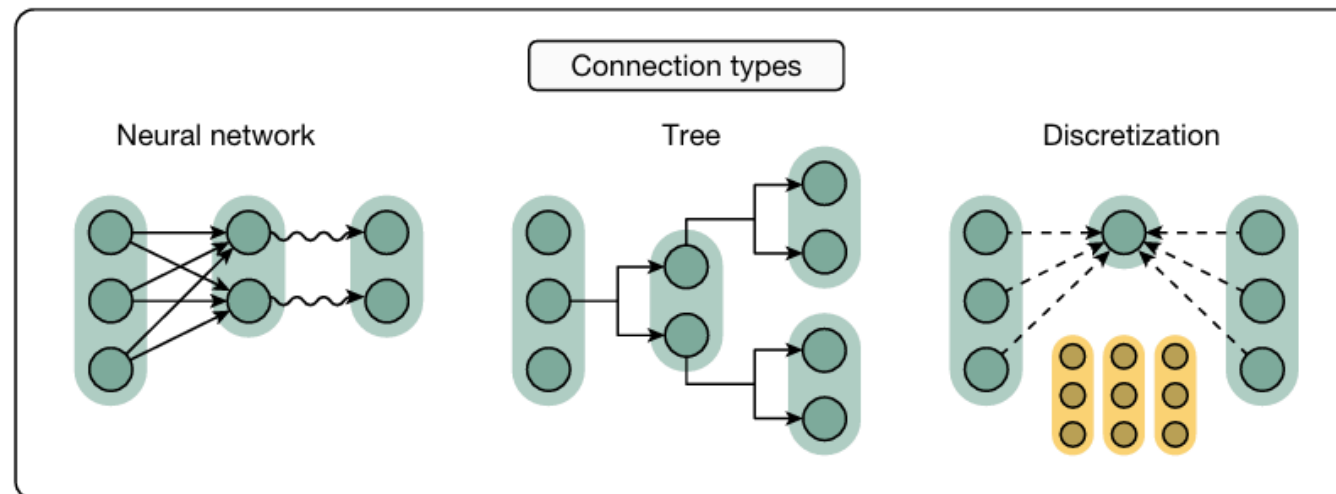
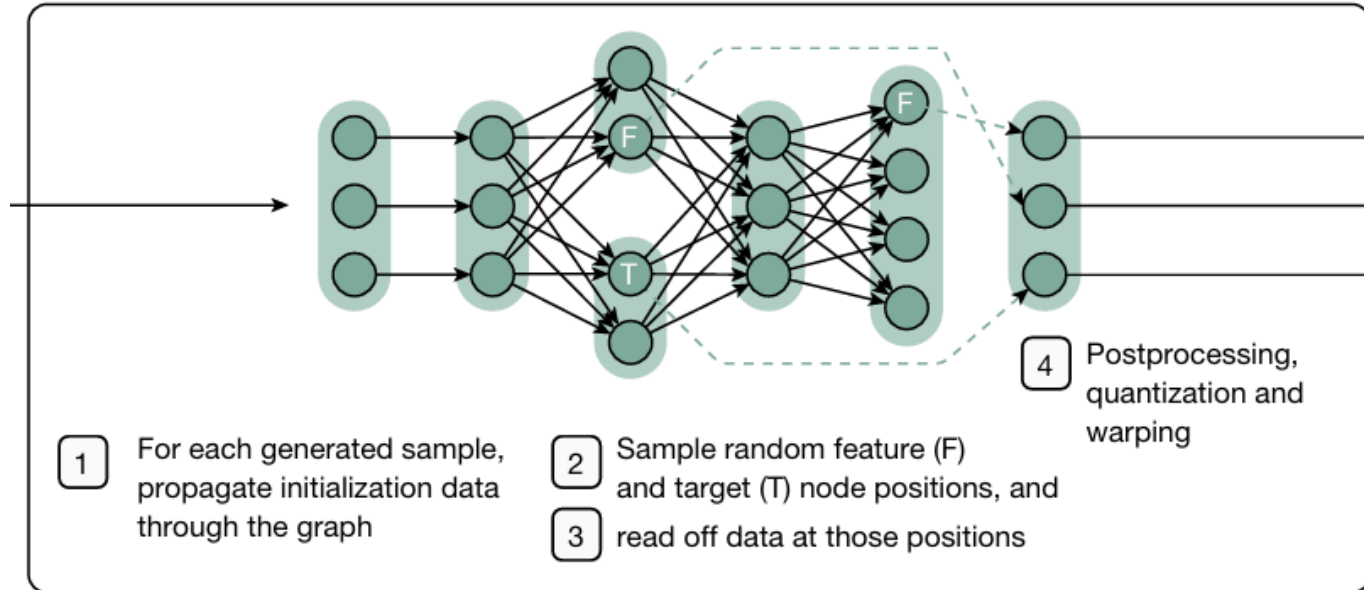
Magpie composition descriptors

- PREN informs the synthetic target; Magpie enriches the input representation.
- The original 17 composition columns are retained.
- Ten descriptors summarize electronegativity, atomic size, melting temperature, and valence-electron properties.
- These descriptors capture alloy-wide chemical patterns relevant to bonding, stability, and passive-film behavior.
- The same transformation is applied during synthetic pretraining and real EPIT evaluation.

21 original features + 10 Magpie descriptors = 31 inputs

Appendix: Synthetic causal model (SCM)

b Build computational graph and graph structure



Appendix: TabICL architecture

TabICL: A Tabular Foundation Model for Large Data

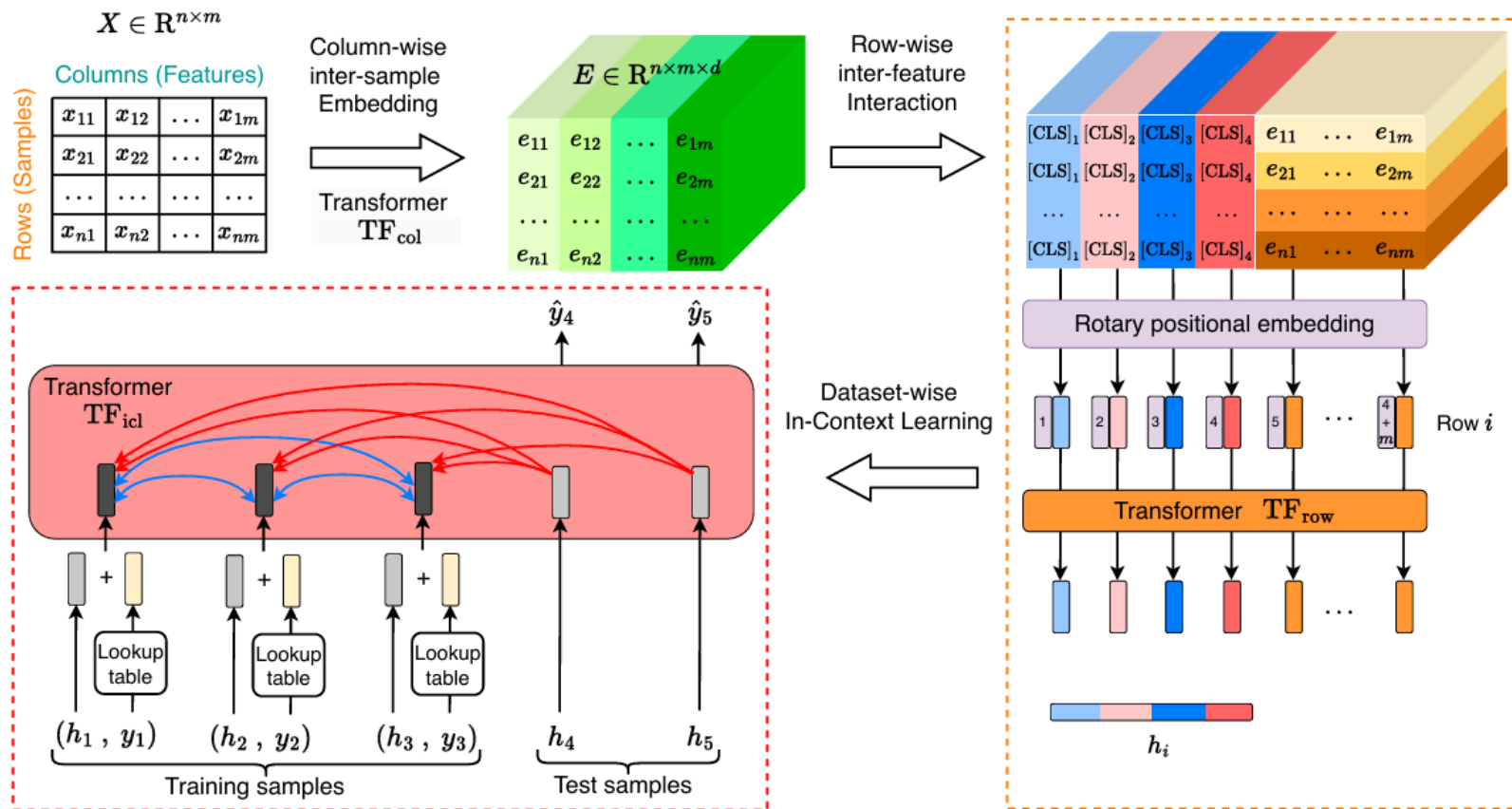


Figure 1. **Architecture Overview of TabICL.** First, column-wise embedding transforms each cell of the input table into an embedding vector using a transformer TF_{col} (Section 3.2), producing E . Next, row-wise interaction prepends four trainable $[CLS]$ tokens to E , applies rotary positional encoding, and processes E row-by-row with a transformer TF_{row} . Concatenating the outputs of $[CLS]$ tokens yields the final row-wise embeddings H . Finally, dataset-wise ICL operates on H and uses a Transformer TF_{icl} to predict the target labels for the test set in a single forward pass. Overall, TabICL consists of three transformers.